

THE SWITCH STATEMENT

The Multi-Option Tool for Clean Logic

CodeHive Academy

Module 7: Structural Control Flow

1. The Switch Expression

A switch statement allows a variable to be tested for equality against a list of values. It is often faster and cleaner than a long list of if-else statements.

The Switch Keyword

The switch keyword is followed by a variable in parentheses. This variable must be an **integral type** (like `int` or `char`).

Brief: Unlike an 'if' that checks conditions, the switch evaluates this variable once and jumps directly to the matching case, making it highly efficient for menu systems and status checks.

2. The Case Keyword

Each possible value you want to check is preceded by the case keyword followed by a colon. This defines the destination of the switch jump.

```
switch (day) { case 1: // Executed if day is 1 // Your code here break;  
case 'A': // Executed if char is 'A' // Your code here break; }
```

Brief: If the switch variable matches the case value, the code following that case executes until it hits a break or the end of the block.

3. The Break Keyword

When C++ reaches a `break` keyword, it "breaks" out of the switch block and proceeds to the code following the closing brace.

Warning: "Fall-through"

Without a `break`, the program will continue to execute every case below the match, regardless of whether they match the variable!

Brief: Use `break` to stop execution as soon as the relevant task is completed. This keeps your CodeHive logic predictable.

4. The Default Keyword

The `default` keyword specifies some code to run if **no case match** is found in the entire block.

```
default: cout << "Value not recognized.";
```

Brief: It acts like the final `else` in an `if-else` chain. While it is not strictly mandatory, it is highly recommended at CodeHive Academy for catching unexpected values and preventing silent bugs.

5. Practical Implementation

Let's look at a complete implementation identifying days of the week.

```
int day = 4; switch (day) { case 1: cout << "Monday"; break; case 4: cout  
<< "Thursday"; // This will execute break; default: cout << "Looking  
forward to the Weekend"; }
```

Architecture Note: Notice how the code is indented inside the switch to make it readable for other developers.

6. Real-Life Subtopic: Menu Selection

On the CodeHive platform, a switch is the industry standard for handling user navigation menus.

- **Case 1:** User opens "Profile".
- **Case 2:** User opens "Compiler".
- **Case 3:** User opens "Settings".
- **Default:** Return to "Home Dashboard".

By using a switch, the interface responds instantly to the integer or character input from the user's click or keystroke.

7. Summary & Quick Reference

Keep this table on your desk while coding your next CodeHive project.

Component	Function	Necessity
Switch(x)	The variable being tested.	Required
Case	Specific value to match.	At least one
Break	Stops code "falling through."	Recommended
Default	Runs if no matches are found.	Optional

